2. Referring to the circuit depicted in Fig. 3.45, count the number of (*a*) nodes; (*b*) elements; (*c*) branches.

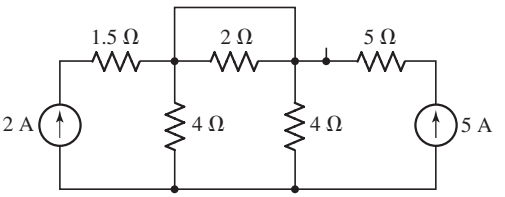


FIGURE 3.45

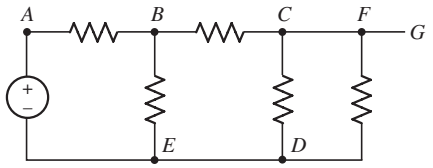


FIGURE 3.46

3. For the circuit of Fig. 3.46:
   (*a*) Count the number of nodes.
   (*b*) In moving from *A* to *B*, have we formed a path? Have we formed a loop?
   (*c*) In moving from *C* to *F* to *G*, have we formed a path? Have we formed a loop?
4. For the circuit of Fig. 3.46:
   (*a*) Count the number of circuit elements.
   (*b*) If we move from *B* to *C* to *D*, have we formed a path? Have we formed a loop?
   (*c*) If we move from *E* to *D* to *C* to *B*, have we formed a path? Have we formed a loop?

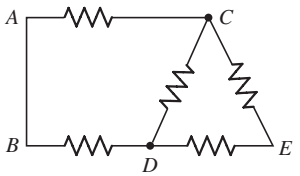


FIGURE 3.47

5. Refer to the circuit of Fig. 3.47, and answer the following:
   (*a*) How many distinct nodes are contained in the circuit?
   (*b*) How many elements are contained in the circuit?
   (*c*) How many branches does the circuit have?
   (*d*) Determine if each of the following represents a path, a loop, both, or neither:
      (*i*) *A* to *B*
      (*ii*) *B* to *D* to *C* to *E*
      (*iii*) *C* to *E* to *D* to *B* to *A* to *C*
      (*iv*) *C* to *D* to *B* to *A* to *C* to *E*

3.2 Kirchhoff's Current Law

6. A local restaurant has a neon sign constructed from 12 separate bulbs; when a bulb fails, it appears as an infinite resistance and cannot conduct current. In wiring the sign, the manufacturer offers two options (Fig. 3.48). From what you've learned about KCL, which one should the restaurant owner select? Explain.

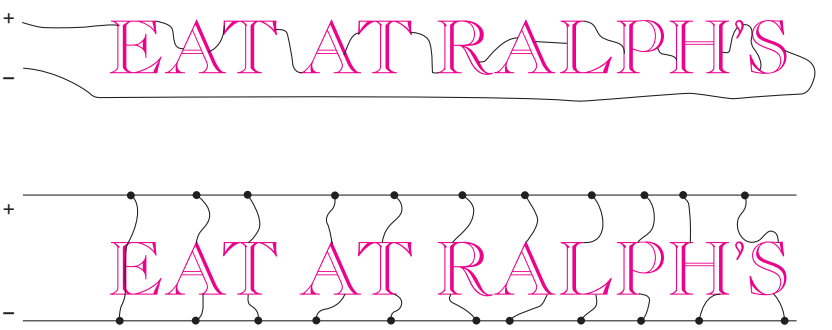


FIGURE 3.48

7. Referring to the single node diagram of Fig. 3.49, compute:
   (*a*) i_B, if $i_A = 1$ A, $i_D = -2$ A, $i_C = 3$ A, and $i_E = 0$;
   (*b*) i_E, if $i_A = -1$ A, $i_B = -1$ A, $i_C = -1$ A, and $i_D = -1$ A.

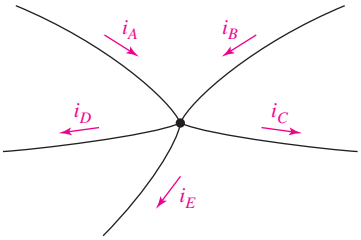


FIGURE 3.49

8. Determine the current labeled I in each of the circuits of Fig. 3.50.

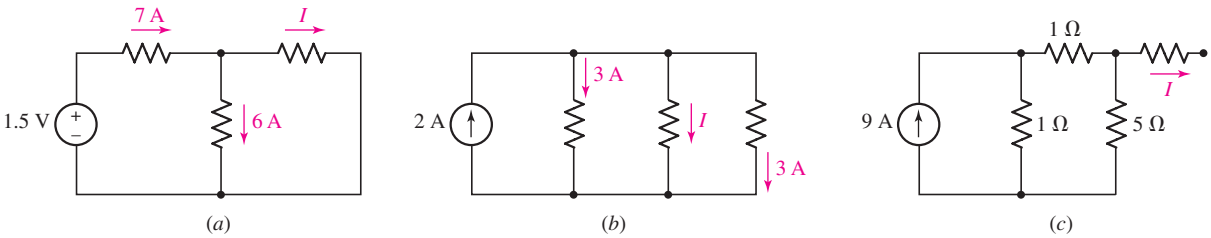


FIGURE 3.50

9. In the circuit shown in Fig. 3.51, the resistor values are unknown, but the 2 V source is known to be supplying a current of 7 A to the rest of the circuit. Calculate the current labeled i_2.

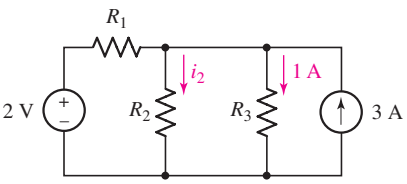


FIGURE 3.51

10. The voltage source in the circuit of Fig. 3.52 has a current of 1 A flowing out of its positive terminal into resistor R_1. Calculate the current labeled i_2.
11. In the circuit depicted in Fig. 3.53, i_x is determined to be 1.5 A, and the 9 V source supplies a current of 7.6 A (that is, a current of 7.6 A leaves the positive reference terminal of the 9 V source). Determine the value of resistor R_A.
12. For the circuit of Fig. 3.54 (which is a model for the dc operation of a bipolar junction transistor biased in forward active region), I_B is measured to be 100 μA. Determine I_C and I_E.

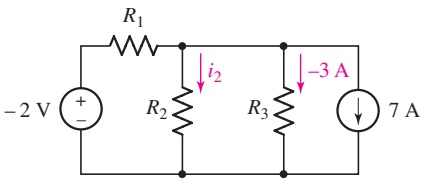


FIGURE 3.52

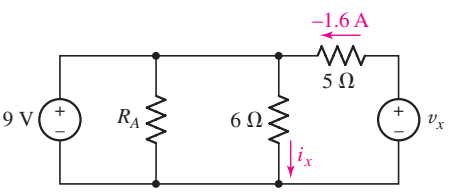


FIGURE 3.53

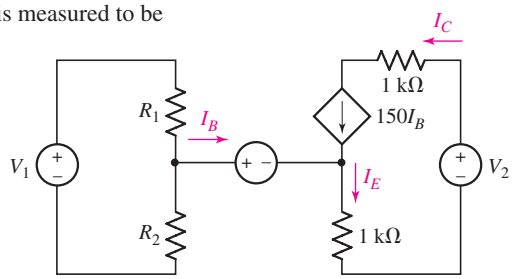


FIGURE 3.54

13. Determine the current labeled I_3 in the circuit of Fig. 3.55.

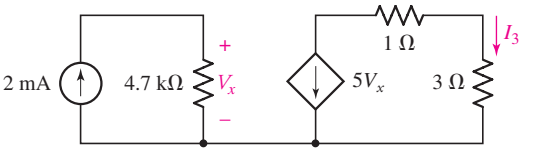


FIGURE 3.55

14. Study the circuit depicted in Fig. 3.56, and explain (in terms of KCL) why the voltage labeled V_x must be zero.

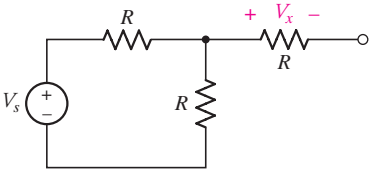


FIGURE 3.56

15. In many households, multiple electrical outlets within a given room are often all part of the same circuit. Draw the circuit for a four-walled room which has a single electrical outlet per wall, with a lamp (represented by a 1 Ω resistor) connected to each outlet.

3.3 Kirchoff's Voltage Law

16. For the circuit of Fig. 3.57:
    (*a*) Determine the voltage v_1 if $v_2 = 0$ and $v_3 = -17$ V.
    (*b*) Determine the voltage v_1 if $v_2 = -2$ V and $v_3 = +2$ V.
    (*c*) Determine the voltage v_2 if $v_1 = 7$ V and $v_3 = 9$ V.
    (*d*) Determine the voltage v_3 if $v_1 = -2.33$ V and $v_2 = -1.70$ V.

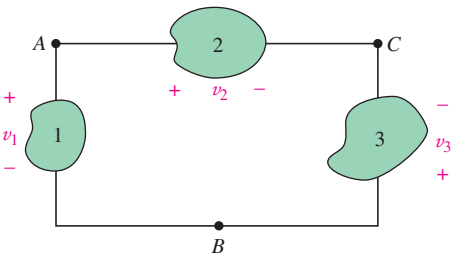


FIGURE 3.57

17. For each of the circuits in Fig. 3.58, determine the voltage v_x and the current i_x.

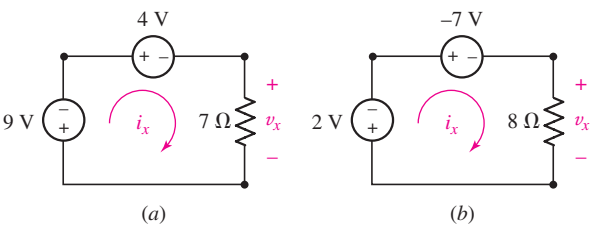


FIGURE 3.58

18. Use KVL to obtain a numerical value for the current labeled i in each circuit depicted in Fig. 3.59.

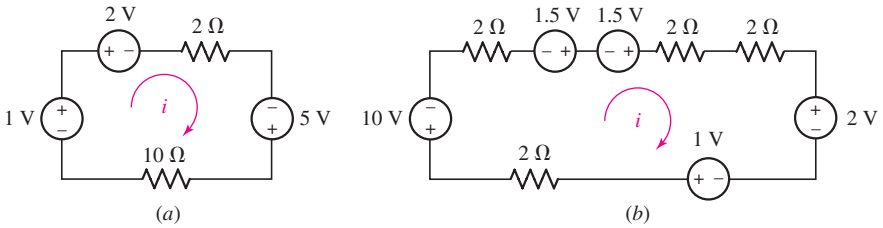


FIGURE 3.59

19. In the circuit of Fig. 3.60, it is determined that $v_1 = 3$ V and $v_3 = 1.5$ V. Calculate v_R and v_2.

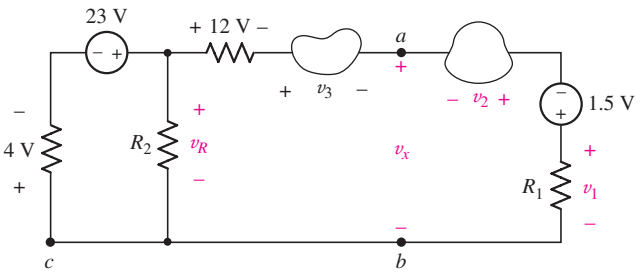


FIGURE 3.60

20. In the circuit of Fig. 3.60, a voltmeter is used to measure the following: $v_1 = 2$ V and $v_3 = -1.5$ V. Calculate v_x.

21. Determine the value of v_x as labeled in the circuit of Fig. 3.61.

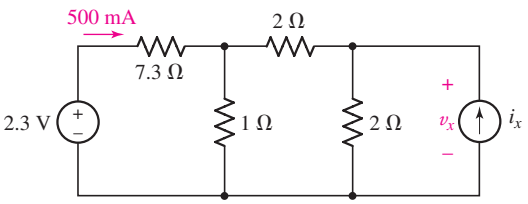


FIGURE 3.61

22. Consider the simple circuit shown in Fig. 3.62. Using KVL, derive the expressions

$$v_1 = v_s \frac{R_1}{R_1 + R_2} \quad \text{and} \quad v_2 = v_s \frac{R_2}{R_1 + R_2}$$

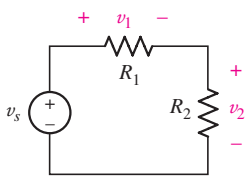


FIGURE 3.62

23. (*a*) Determine a numerical value for each current and voltage (i_1, v_1, etc.) in the circuit of Fig. 3.63. (*b*) Calculate the power absorbed by each element and verify that they sum to zero.

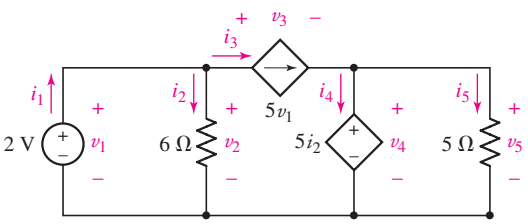


FIGURE 3.63

24. The circuit shown in Fig. 3.64 includes a device known as an op amp. This device has two unusual properties in the circuit shown: (1) $V_d = 0$ V, and (2) no current can flow into either input terminal (marked "−" and "+" inside the symbol), but it *can* flow through the output terminal (marked "OUT"). This seemingly impossible situation—in direct conflict with KCL—is a result of power leads to the device that are not included in the symbol. Based on this information, calculate V_{out}. (*Hint*: two KVL equations are required, both involving the 5 V source.)

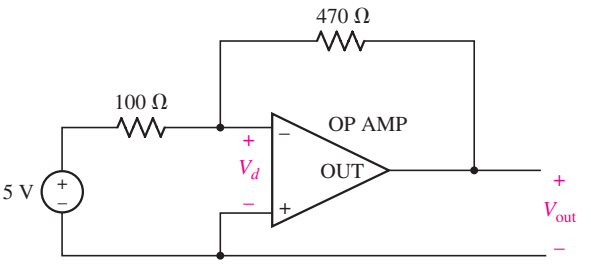


FIGURE 3.64

3.4 The Single-Loop Circuit

25. The circuit of Fig. 3.12*b* is constructed with the following: $v_{s1} = -8$ V, $R_1 = 1\ \Omega$, $v_{s2} = 16$ V, and $R_2 = 4.7\ \Omega$. Calculate the power absorbed by each element. Verify that the absorbed powers sum to zero.

26. Obtain a numerical value for the power absorbed by each element in the circuit shown in Fig. 3.65.

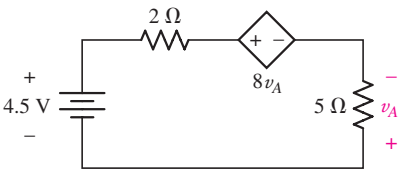


FIGURE 3.65

27. Compute the power absorbed by each element of the circuit of Fig. 3.66.

28. Compute the power absorbed by each element in the circuit of Fig. 3.67 if the mysterious element X is (*a*) a 13 Ω resistor; (*b*) a dependent voltage source labeled $4v_1$, "+" reference on top; (*c*) a dependent voltage source labeled $4i_x$, "+" reference on top.

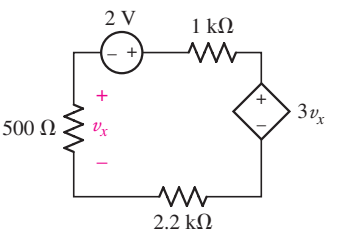


FIGURE 3.66

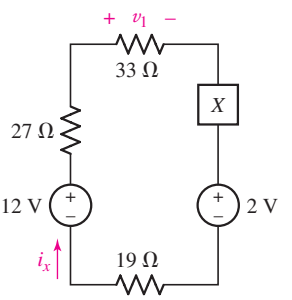


FIGURE 3.67

29. Kirchhoff's laws apply whether or not Ohm's law applies to a particular element. The *I-V* characteristic of a diode, for example, is given by

$$I_D = I_S \left(e^{V_D/V_T} - 1\right)$$

where $V_T = 27$ mV at room temperature and I_S can vary from 10^{-12} to 10^{-3} A. In the circuit of Fig. 3.68, use KVL/KCL to obtain V_D if $I_S = 29$ pA. (*Note: This problem results in a transcendental equation, requiring an iterative approach to obtaining a numerical solution. Most scientific calculators will perform such a function.*)

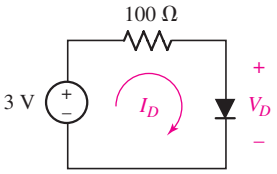


FIGURE 3.68

3.5 The Single-Node-Pair Circuit

30. Referring to the circuit of Fig. 3.69, (*a*) determine the two currents i_1 and i_2; (*b*) compute the power absorbed by each element.

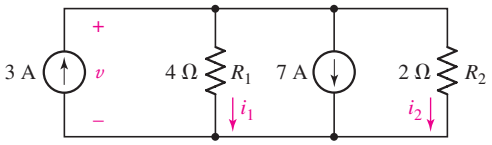


FIGURE 3.69

31. Determine a value for the voltage v as labeled in the circuit of Fig. 3.70, and compute the power supplied by the two current sources.

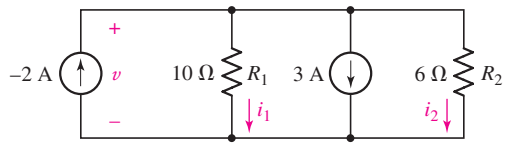


FIGURE 3.70

32. Referring to the circuit depicted in Fig. 3.71, determine the value of the voltage v.

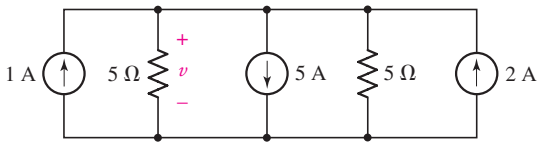


FIGURE 3.71

33. Determine the voltage v as labeled in Fig. 3.72, and calculate the power supplied by each current source.

34. Although drawn so that it may not appear obvious at first glance, the circuit of Fig. 3.73 is in fact a single-node-pair circuit. (*a*) Determine the power absorbed by each resistor. (*b*) Determine the power supplied by each current source. (*c*) Show that the sum of the absorbed power calculated in (*a*) is equal to the sum of the supplied power calculated in (*c*).

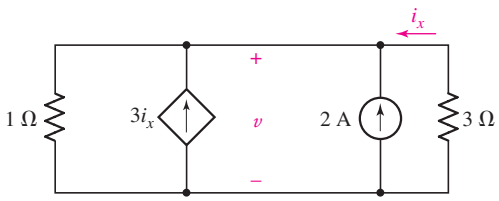


FIGURE 3.72

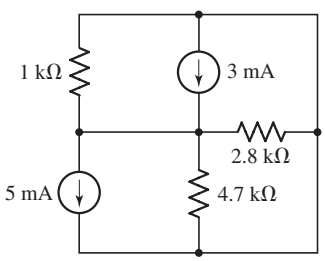


FIGURE 3.73

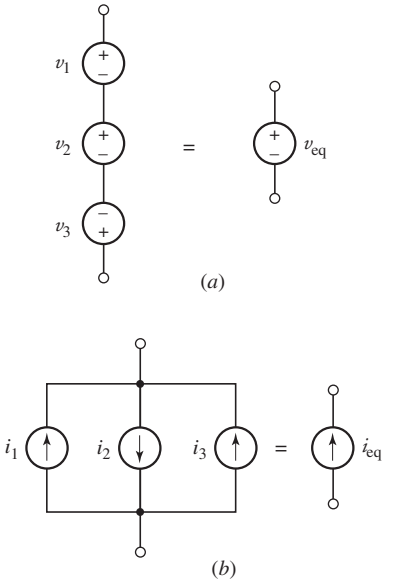


FIGURE 3.74

3.6 Series and Parallel Connected Sources

35. Determine the numerical value for v_{eq} in Fig. 3.74*a*, if (*a*) $v_1 = 0$, $v_2 = -3$ V, and $v_3 = +3$ V; (*b*) $v_1 = v_2 = v_3 = 1$ V; (*c*) $v_1 = -9$ V, $v_2 = 4.5$ V, $v_3 = 1$ V.

36. Determine the numerical value for i_{eq} in Fig. 3.74*b*, if (*a*) $i_1 = 0$, $i_2 = -3$ A, and $i_3 = +3$ A; (*b*) $i_1 = i_2 = i_3 = 1$ A; (*c*) $i_1 = -9$ A, $i_2 = 4.5$ A, $i_3 = 1$ A.

37. For the circuit presented in Fig. 3.75, determine the current labeled i by first combining the four sources into a single equivalent source.

38. Determine the value of v_1 required to obtain a zero value for the current labeled i in the circuit of Fig. 3.76.

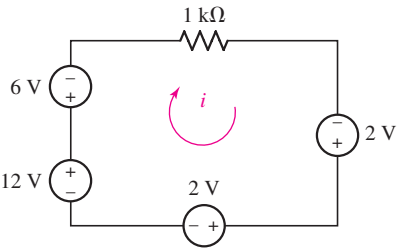


FIGURE 3.75

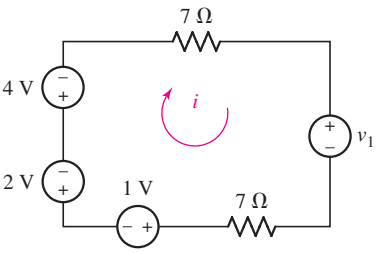


FIGURE 3.76

39. (*a*) For the circuit of Fig. 3.77, determine the value for the voltage labeled v, after first simplifying the circuit to a single current source in parallel with two resistors. (*b*) Verify that the power supplied by your equivalent source is equal to the sum of the supplied powers of the individual sources in the original circuit.

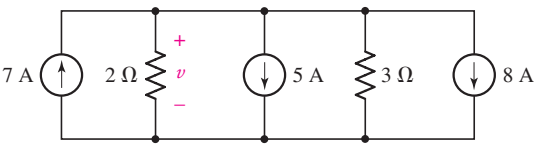


FIGURE 3.77

40. What value of I_S in the circuit of Fig. 3.78 will result in a zero voltage v?

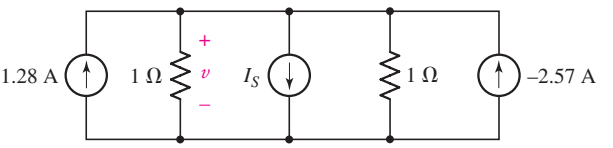


FIGURE 3.78

41. (*a*) Determine the values for I_X and V_Y in the circuit shown in Fig. 3.79. (*b*) Are those values necessarily unique for that circuit? Explain. (*c*) Simplify the circuit of Fig. 3.79 as much as possible and still maintain the values for v and i. (Your circuit must contain the 1 Ω resistor.)

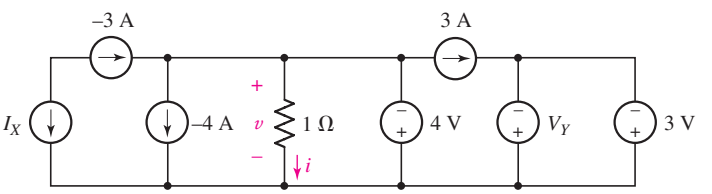


FIGURE 3.79

3.7 Resistors in Series and Parallel

42. Determine the equivalent resistance of each of the networks shown in Fig. 3.80.

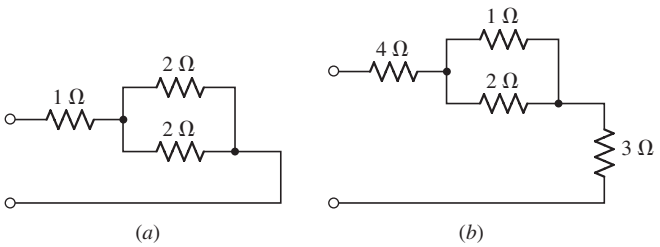


FIGURE 3.80

43. For each network depicted in Fig. 3.81, determine a single equivalent resistance.

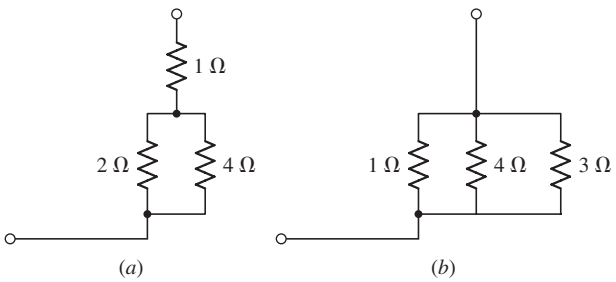


FIGURE 3.81

44. (*a*) Simplify the circuit of Fig. 3.82 as much as possible by using source and resistor combinations. (*b*) Calculate i, using your simplified circuit. (*c*) To what voltage should the 1 V source be changed to reduce i to zero? (*d*) Calculate the power absorbed by the 5 Ω resistor.

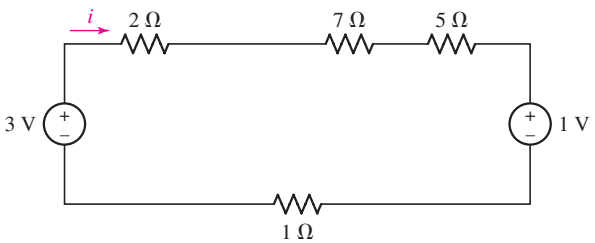


FIGURE 3.82

45. (*a*) Simplify the circuit of Fig. 3.83, using appropriate source and resistor combinations. (*b*) Determine the voltage labeled v, using your simplified circuit. (*c*) Calculate the power provided by the 2 A source to the rest of the circuit.

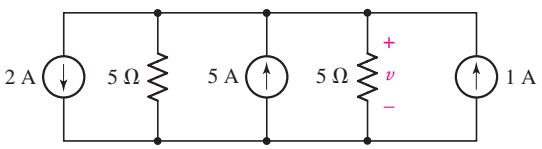


FIGURE 3.83

46. Making appropriate use of resistor combination techniques, calculate i_3 in the circuit of Fig. 3.84 and the power provided to the circuit by the single current source.

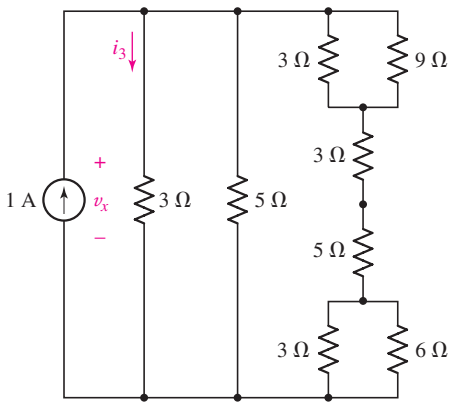


FIGURE 3.84

47. Calculate the voltage labeled v_x in the circuit of Fig. 3.85 after first simplifying, using appropriate source and resistor combinations.

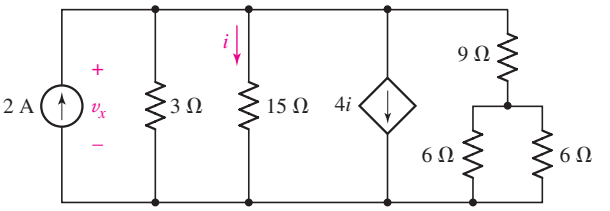


FIGURE 3.85

48. Determine the power absorbed by the 15 Ω resistor in the circuit of Fig. 3.86.

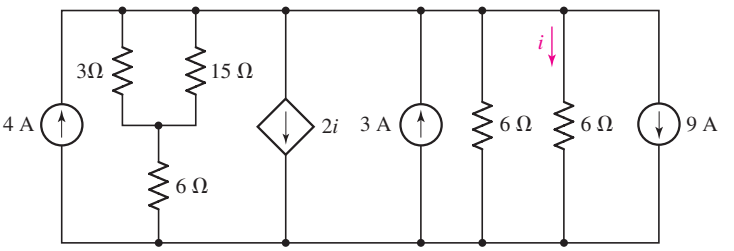


FIGURE 3.86

49. Calculate the equivalent resistance R_{eq} of the network shown in Fig. 3.87 if $R_1 = 2R_2 = 3R_3 = 4R_4$ etc. and $R_{11} = 3\ \Omega$.

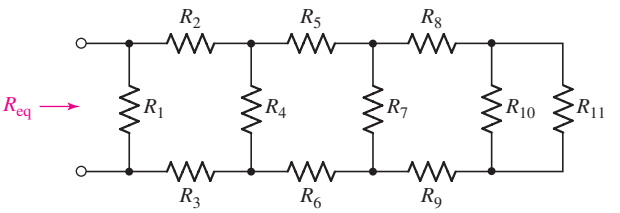


FIGURE 3.87

50. Show how to combine four 100 Ω resistors to obtain an equivalent resistance of (*a*) 25 Ω; (*b*) 60 Ω; (*c*) 40 Ω.

3.8 Voltage and Current Division

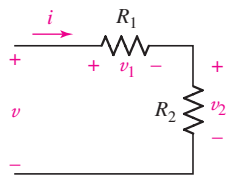


FIGURE 3.88

51. In the voltage divider network of Fig. 3.88, calculate (*a*) v_2 if $v = 9.2$ V and $v_1 = 3$ V; (*b*) v_1 if $v_2 = 1$ V and $v = 2$ V; (*c*) v if $v_1 = 3$ V and $v_2 = 6$ V; (*d*) R_1/R_2 if $v_1 = v_2$; (*e*) v_2 if $v = 3.5$ V and $R_1 = 2R_2$; (*f*) v_1 if $v = 1.8$ V, $R_1 = 1$ kΩ, and $R_2 = 4.7$ kΩ.

52. In the current divider network represented in Fig. 3.89, calculate (*a*) i_1 if $i = 8$ A and $i_2 = 1$ A; (*b*) v if $R_1 = 100$ kΩ, $R_2 = 100$ kΩ, and $i = 1$ mA; (*c*) i_2 if $i = 20$ mA, $R_1 = 1$ Ω, and $R_2 = 4$ Ω; (*d*) i_1 if $i = 10$ A, $R_1 = R_2 = 9$ Ω; (*e*) i_2 if $i = 10$ A, $R_1 = 100$ MΩ, and $R_2 = 1$Ω.

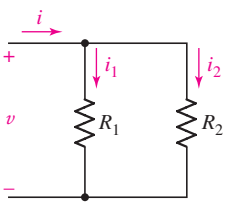


FIGURE 3.89

53. Choose a voltage $v < 2.5$ V and values for the resistors R_1, R_2, R_3, and R_4 in the circuit of Fig. 3.90 so that $i_1 = 1$ A, $i_2 = 1.2$ A, $i_3 = 8$ A, and $i_4 = 3.1$ A.

54. Employ voltage division to assist in the calculation of the voltage labeled v_x in the circuit of Fig. 3.91.

55. A network is constructed from a series connection of five resistors having values 1 Ω, 3 Ω, 5 Ω, 7 Ω, and 9 Ω. If 9 V is connected across the terminals of the network, employ voltage division to calculate the voltage across the 3 Ω resistor, and the voltage across the 7 Ω resistor.

56. Employing resistance combination and current division as appropriate, determine values for i_1, i_2, and v_3 in the circuit of Fig. 3.92.

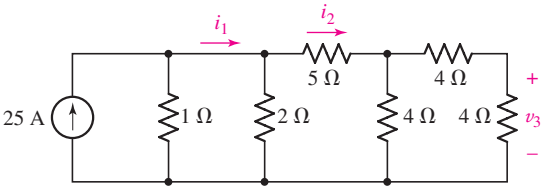


FIGURE 3.92

57. In the circuit of Fig. 3.93, only the voltage v_x is of interest. Simplify the circuit using appropriate resistor combinations and iteratively employ voltage division to determine v_x.

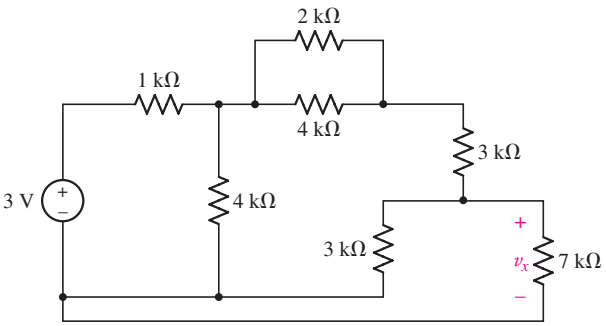


FIGURE 3.93

Chapter-Integrating Exercises

58. The circuit shown in Fig. 3.94 is a linear model of a bipolar junction transistor biased in the forward active region of operation. Explain why voltage division is not a valid approach for determining the voltage across either 10 kΩ resistor.

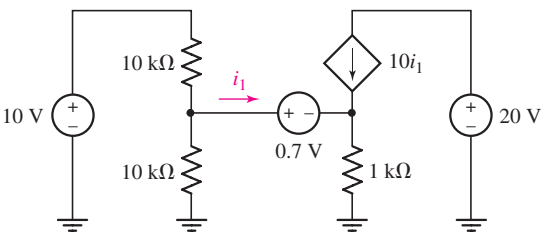


FIGURE 3.94

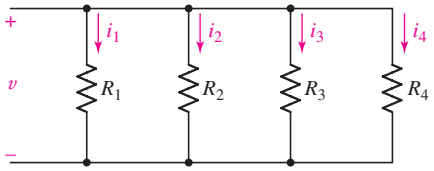


FIGURE 3.90

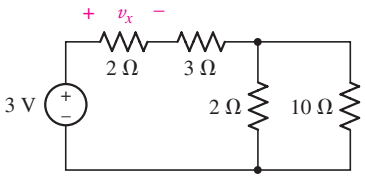


FIGURE 3.91

59. A common midfrequency model for a field effect–based amplifier circuit is shown in Fig. 3.95. If the controlling parameter g_m (known as the *transconductance*) is equal to 1.2 mS, employ current division to obtain the current through the 1 kΩ resistor, and then calculate the amplifier output voltage v_{out}.

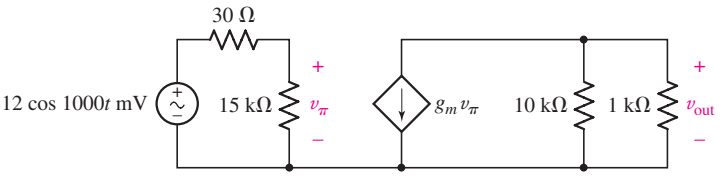


FIGURE 3.95

60. The circuit depicted in Fig. 3.96 is routinely employed to model the midfrequency operation of a bipolar junction transistor–based amplifier. Calculate the amplifier output v_{out} if the transconductance g_m is equal to 322 mS.

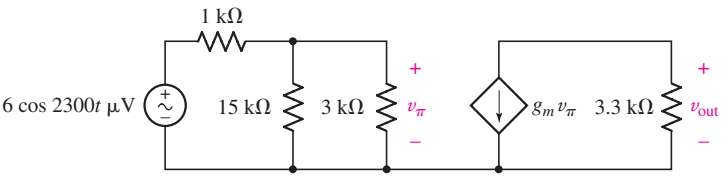


FIGURE 3.96

61. With regard to the circuit shown in Fig. 3.97, compute (*a*) the voltage across the two 10 Ω resistors, assuming the top terminal is the positive reference; (*b*) the power dissipated by the 4 Ω resistor.

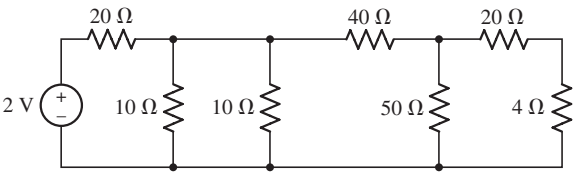


FIGURE 3.97

62. Delete the leftmost 10 Ω resistor in the circuit of Fig. 3.97, and compute (*a*) the current flowing into the left-hand terminal of the 40 Ω resistor; (*b*) the power supplied by the 2 V source; (*c*) the power dissipated by the 4 Ω resistor.

63. Consider the seven-element circuit depicted in Fig. 3.98. (*a*) How many nodes, loops, and branches does it contain? (*b*) Calculate the current flowing through each resistor. (*c*) Determine the voltage across the current source, assuming the top terminal is the positive reference terminal.

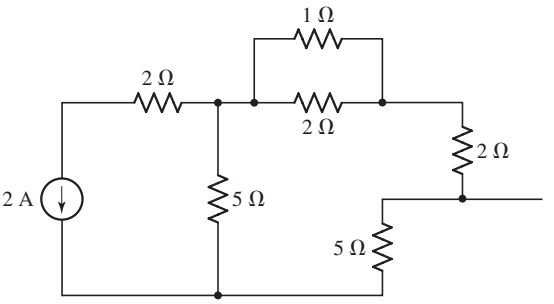


FIGURE 3.98